HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Graphical User Interfaces (GUIs)** for scientific instrumentation using **Java Swing**, integrating real-time data streams and control logic.
- Engineered and deployed **data acquisition solutions** in Python, processing instrument readings and generating scheduled reports for system performance analysis.
- Implemented **instrument control interfaces** using **TCP/IP and MQTT** protocols, ensuring reliable communication between the central control system and distributed hardware components.
- Resolved complex **hardware, software, and firmware issues** through systematic debugging and root cause analysis, restoring system functionality within defined SLAs.
- Collaborated with multidisciplinary teams, including hardware engineers and domain experts, to define requirements and integrate control solutions for high-performance instruments.
- Contributed to the continuous improvement of the software architecture by participating in **code reviews** and suggesting enhancements for maintainability and scalability.
- Containerised instrument control applications using **Docker** and deployed them to **AWS EC2 instances**, streamlining the deployment process and ensuring environment consistency.

SKILLS

Languages – Python, Java, SQL, JavaScript, C#

Frameworks/Libraries – Spring Boot, Java Swing, MQTT, TCP/IP, pandas, NumPy, scikit-learn

Tools & Technologies – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

Databases – PostgreSQL, MySQL, Oracle

Protocols – TCP/IP, MQTT

Development Methodologies – Agile, Scrum

PROJECTS

Instrument Control System Development (HCL Technologies)

- Developed a core **Java Spring Boot application** to manage instrument states and orchestrate complex operational sequences for scientific equipment.
- Integrated **REST APIs** for seamless data exchange between the control system and external monitoring platforms, enhancing real-time visibility.
- Wrote comprehensive unit tests using **JUnit**, achieving over 85% code coverage and ensuring the stability of critical control logic.

Automated Reporting & Data Ingestion (HCL Technologies)

- Created Python scripts to automate the extraction, transformation, and loading (ETL) of data from various instrument logs and databases into a central repository.
- Built **data acquisition pipelines** that processed approximately 5,000 data points per minute, enabling faster analysis of system performance trends.
- Designed and implemented **SQL queries** for data validation and reporting, supporting operational decision-making for a financial-services dashboard.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)